

Differential Equations of First Order

11.1. DEFINITIONS

(i) A **differential equation** is an equation involving differentials or differential coefficients.

$$\text{Thus,} \quad \frac{dy}{dx} = x^2 - 1 \quad \dots(1)$$

$$\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 + y = 0 \quad \dots(2)$$

$$\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} \cdot \frac{dy}{dx} + x^2\left(\frac{dy}{dx}\right)^3 = 0 \quad \dots(3)$$

$$y = x \frac{dy}{dx} + \frac{c}{\frac{dy}{dx}} \quad \dots(4)$$

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = k \frac{d^2y}{dx^2} \quad \dots(5)$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu \quad \dots(6)$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = x + y \quad \dots(7)$$

are all differential equations.

(ii) Differential equations which involve only one independent variable and the differential co-efficients with respect to it are called **ordinary differential equations**.

Thus equations (1) to (5) are all ordinary differential equations.

(iii) Differential equations which involve two or more independent variables and partial derivatives with respect to them are called **partial differential equations**.

Thus equations (6) and (7) are partial differential equations.

(iv) The **order** of a differential equation is the order of the highest ordered derivative occurring in the differential equation.

Thus equations (1) and (4) are of first order ; equations (2) and (5) are of the second order while equation (3) is of the third order.

(v) The **degree** of a differential equation is the degree of the highest ordered derivative which occurs in it provided the equation has been made free of the radical signs and fractional powers as far as the derivatives are concerned.

Thus, equations (1), (2) and (3) are of the first degree.

Equation (4) is $y \frac{dy}{dx} = x \left(\frac{dy}{dx} \right)^2 + c$. It is of the second degree.

Equation (5) is $\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^3 = k^2 \left(\frac{d^2y}{dx^2} \right)^2$. It is of the second degree.

(vi) **Solution of a Differential Equation.** A solution (or integral) of a differential equation is a relation, free from derivatives, between the variables which satisfies the given equation.

Thus if $y = f(x)$ be the solution, then by replacing y and its derivatives with respect to x , the given differential equation will reduce to an identity.

For example, $y = c_1 \cos x + c_2 \sin x$ is the solution of the differential equation

$$\frac{d^2y}{dx^2} + y = 0$$

Since $\frac{dy}{dx} = -c_1 \sin x + c_2 \cos x$

$$\frac{d^2y}{dx^2} = -c_1 \cos x - c_2 \sin x = -y.$$

The **general** (or **complete**) **solution** of a differential equation is that in which the number of independent arbitrary constants is equal to the order of the differential equation.

Thus, $y = c_1 \cos x + c_2 \sin x$ (involving two arbitrary constants c_1, c_2) is the general solution of the differential equation $\frac{d^2y}{dx^2} + y = 0$ of second order.

A **particular solution** of a differential equation is that which is obtained from its general solution by giving particular values to the arbitrary constants.

For example, $y = c_1 e^x + c_2 e^{-x}$ is the general solution of the differential equation $\frac{d^2y}{dx^2} - y = 0$, whereas $y = e^x - e^{-x}$ or $y = e^x$ are its particular solutions.

The solution of a differential equation of n^{th} order is its particular solution if it contains less than n arbitrary constants.

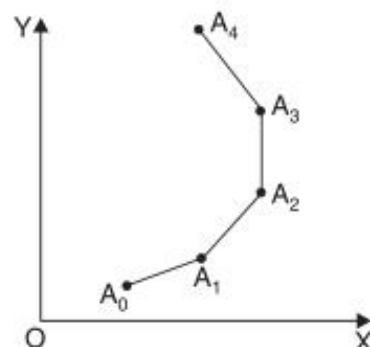
11.2. GEOMETRICAL MEANING OF A DIFFERENTIAL EQUATION OF THE FIRST ORDER AND FIRST DEGREE

Let $f\left(x, y, \frac{dy}{dx}\right) = 0$... (1)

be a differential equation of the first order and first degree.

We know that the direction of a curve at a particular point is determined by drawing a tangent line at that point, *i.e.*, its slope is given by $\frac{dy}{dx}$ at that particular point.

Let $A_0 (x_0, y_0)$ be any point in the plane. Let $m_0 = \frac{dy_0}{dx_0}$ be the slope of the curve at A_0 derived from (1). Take a neighbouring point $A_1 (x_1, y_1)$ such that the slope of A_0A_1 is m_0 . Let $m_1 = \frac{dy_1}{dx_1}$ be the slope of the curve at A_1 derived from (1). Take a neighbouring point $A_2 (x_2, y_2)$ such that the slope of A_1A_2 is m_1 . Continuing like this, we get a succession of points. If the points are taken sufficiently close to each other, they approximate a smooth curve $C : y = \phi(x)$ which is a solution of (1) corresponding to the initial point $A_0 (x_0, y_0)$. Any point on C and the slope of the tangent at that point satisfy (1). If the moving point starts at any other point, not on C and moves as before, it will describe another curve. The equation of each such curve is a *particular solution* of the differential equation (1). The equation of the system of all such curves is the *general solution* of (1).



11.3. FORMATION OF A DIFFERENTIAL EQUATION

Differential equations are formed by elimination of arbitrary constants. To eliminate two arbitrary constants, we require two more equations besides the given relation, leading us to second order derivatives and hence a differential equation of the second order. Elimination of n arbitrary constants leads us to n^{th} order derivatives and hence a differential equation of the n^{th} order.

ILLUSTRATIVE EXAMPLES

Example 1. Eliminate the constants from the equation $y = e^x (A \cos x + B \sin x)$ and obtain the differential equation.

Sol. There are two arbitrary constants A and B in the given equation.

Differentiating the given eqn. w.r.t. x , we have

$$\begin{aligned} \frac{dy}{dx} &= e^x (A \cos x + B \sin x) + e^x (-A \sin x + B \cos x) \\ &= y + e^x (-A \sin x + B \cos x) \end{aligned} \quad \dots(1)$$

Differentiating again w.r.t. x , we have

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{dy}{dx} + e^x (-A \sin x + B \cos x) + e^x (-A \cos x - B \sin x) \\ &= \frac{dy}{dx} + \left(\frac{dy}{dx} - y \right) - y \\ \Rightarrow \quad \frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y &= 0 \end{aligned}$$

Example 2. Find the differential equation of all circles touching the axis of y at the origin and centres on the axis of x .

Sol. The equation of such a circle is

$$(x - h)^2 + y^2 = h^2$$

or
$$x^2 + y^2 - 2hx = 0$$

where h is the only arbitrary constant.

Differentiating (1) w.r.t. x , we have

$$2x + 2y \frac{dy}{dx} - 2h = 0$$

or
$$h = x + y \frac{dy}{dx}$$

Substituting the value of h in (1), we have

$$x^2 + y^2 - 2x \left(x + y \frac{dy}{dx} \right) = 0$$

or
$$2xy \frac{dy}{dx} + x^2 - y^2 = 0$$

Example 3. Form the differential equation of all circles of radius a .

Sol. The equation of any circle of radius a is $(x - h)^2 + (y - k)^2 = a^2$... (1)

where (h, k) the coordinates of the centre are arbitrary.

Differentiating (1) w.r.t. x , we have $2(x - h) + 2(y - k) \frac{dy}{dx} = 0$

or
$$(x - h) + (y - k) \frac{dy}{dx} = 0$$
 ... (2)

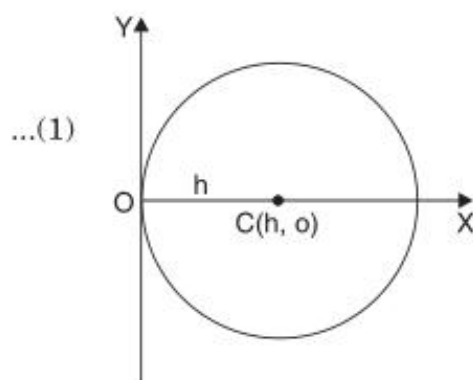
Differentiating again, we have $1 + (y - k) \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 0$... (3)

From (3),
$$y - k = - \frac{1 + \left(\frac{dy}{dx} \right)^2}{\frac{d^2y}{dx^2}}$$

and from (2),
$$x - h = - (y - k) \frac{dy}{dx} = \frac{\frac{dy}{dx} \left[1 + \left(\frac{dy}{dx} \right)^2 \right]}{\frac{d^2y}{dx^2}}$$

Substituting the values of $(x - h)$ and $(y - k)$ in (1), we get

$$\frac{\left(\frac{dy}{dx} \right)^2 \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^2}{\left(\frac{d^2y}{dx^2} \right)^2} + \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^2}{\left(\frac{d^2y}{dx^2} \right)^2} = a^2$$



or

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = a^2 \left(\frac{d^2y}{dx^2}\right)^2.$$

TEST YOUR KNOWLEDGE

Eliminate the arbitrary constants and obtain the differential equations :

- | | | |
|--------------------------------|-------------------------------|------------------------------------|
| 1. $y = cx + c^2$ | 2. $y = A + Bx + Cx^2$ | 3. $y = A \cos 2t + B \sin 2t$ |
| 4. $y = Ae^{3x} + Be^{2x}$ | 5. $y = Ae^x + Be^{-x} + C$ | 6. $y = ax^3 + bx^2$ |
| 7. $xy = Ae^x + Be^{-x} + x^2$ | 8. $x = A \cos (nt + \alpha)$ | 9. $y = ae^{2x} + be^{-3x} + ce^x$ |
| 10. $Ax^2 + By^2 = 1$ | 11. $y^2 - 2ay + x^2 = a^2$ | 12. $e^{2y} + 2ax e^y + a^2 = 0$ |

Find the differential equations of :

- All straight lines in a plane.
- All circles of radius r whose centres lie on the x -axis.
- All parabolas with x -axis as the axis and $(a, 0)$ as focus.
- All conics whose axes coincide with the axes of co-ordinates.
- All circle in a plane.
- All circles in the first quadrant which touch the coordinate axes.
- All circles passing through the origin and having centres on the x -axis.
- All parabolas with latus rectum ' $4a$ ' and axis parallel to the x -axis.

Answers

- | | | |
|--|--|---|
| 1. $x \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2 = y$ | 2. $\frac{d^3y}{dx^3} = 0$ | 3. $\frac{d^2y}{dt^2} + 4y = 0$ |
| 4. $\frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$ | 5. $\frac{d^3y}{dx^3} - \frac{dy}{dx} = 0$ | 6. $x^2 \frac{d^2y}{dx^2} - 4x \frac{dy}{dx} + 6y = 0$ |
| 7. $x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + x^2 - xy - 2 = 0$ | 8. $\frac{d^2x}{dt^2} + n^2x = 0$ | 9. $\frac{d^3y}{dx^3} - 7 \frac{dy}{dx} + 6y = 0$ |
| 10. $xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx}\right)^2 - y \frac{dy}{dx} = 0$ | 11. $(x^2 - 2y^2) \left(\frac{dy}{dx}\right)^2 - 4xy \frac{dy}{dx} - x^2 = 0$ | |
| 12. $(1 - x^2) \left(\frac{dy}{dx}\right)^2 + 1 = 0$ | 13. $\frac{d^2y}{dx^2} = 0$ | 14. $y^2 \left[1 + \left(\frac{dy}{dx}\right)^2\right] = r^2$ |
| 15. $y \frac{dy}{dx} = 2a$ | 16. $xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx}\right)^2 = y \frac{dy}{dx}$ | |
| 17. $\left[1 + \left(\frac{dy}{dx}\right)^2\right] \frac{d^3y}{dx^3} - 3 \frac{dy}{dx} \left(\frac{d^2y}{dx^2}\right)^2 = 0$ | 18. $(x - y)^2 \left[1 + \left(\frac{dy}{dx}\right)^2\right] = \left(x + y \frac{dy}{dx}\right)^2$ | |
| 19. $x^2 - y^2 + 2xy \frac{dy}{dx} = 0$ | 20. $2a \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 0.$ | |

11.4. SOLUTION OF DIFFERENTIAL EQUATIONS OF THE FIRST ORDER AND FIRST DEGREE

All differential equations of the first order and first degree cannot be solved. Only those among them which belong to (or can be reduced to) one of the following categories can be solved by the standard methods.

- (i) Equations in which variables are separable.
 (ii) Homogeneous equations. (iii) Linear equations. (iv) Exact equations.

11.5. VARIABLES SEPARABLE FORM

If a differential equation of the first order and first degree can be put in the form where dx and all terms containing x are at one place, also dy and all terms containing y are at one place, then the variables are said to be separable.

Thus the general form of such an equation is $f(x) dx + \phi(y) dy = 0$

Integrating, we get $\int f(x) dx + \int \phi(y) dy = c$ which is the general solution, c being an arbitrary constant.

Note 1. Any equation of the form $f_1(x) \phi_2(y) dx + f_2(x) \phi_1(y) dy = 0$ can be expressed in the above form by dividing throughout by $f_2(x) \phi_2(y)$.

Thus
$$\frac{f_1(x)}{f_2(x)} dx + \frac{\phi_1(y)}{\phi_2(y)} dy = 0 \quad \text{or} \quad f(x) dx + \phi(y) dy = 0.$$

Note 2. Differential equations of the form $\frac{dy}{dx} = f(ax + by + c)$... (1)

can be reduced to a form in which the variables are separable by the substitution $ax + by + c = t$

so that
$$a + b \frac{dy}{dx} = \frac{dt}{dx} \quad \text{or} \quad \frac{dy}{dx} = \frac{1}{b} \left(\frac{dt}{dx} - a \right)$$

$\therefore$ Equation (1) becomes $\frac{1}{b} \left(\frac{dt}{dx} - a \right) = f(t) \quad \text{or} \quad \frac{dt}{dx} = a + bf(t).$

or
$$\frac{dt}{a + bf(t)} = dx$$

After integrating both sides, t is to be replaced by its value.

ILLUSTRATIVE EXAMPLES

Example 1. Solve $\frac{dy}{dx} = \frac{x(2 \log x + 1)}{\sin y + y \cos y}.$

Sol. The given equation can be written as $(\sin y + y \cos y) dy = x(2 \log x + 1) dx$

Integrating both sides $\int (\sin y + y \cos y) dy = \int x(2 \log x + 1) dx + c$

$\Rightarrow -\cos y + y \sin y - \int 1 \cdot \sin y dy = 2 \int (\log x) \cdot x dx + \int x dx + c$

$\Rightarrow -\cos y + y \sin y + \cos y = 2 \left[(\log x) \cdot \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx \right] + \frac{x^2}{2} + c$

$\Rightarrow y \sin y = x^2 \log x + c \quad | \quad c \text{ is an arbitrary constant}$

which is the general solution of the given equation.

Example 2. Solve $3e^x \tan y \, dx + (1 + e^x) \sec^2 y \, dy = 0$, given $y = \frac{\pi}{4}$ when $x = 0$.

Sol. The given equation can be written as $\frac{3e^x}{1+e^x} dx + \frac{\sec^2 y}{\tan y} dy = 0$

Integrating, we have $3 \log(1 + e^x) + \log \tan y = \log c$

$$\Rightarrow \log(1 + e^x)^3 \tan y = \log c$$

$$\Rightarrow (1 + e^x)^3 \tan y = c \quad \dots(1)$$

which is the general solution of the given equation.

Since $y = \frac{\pi}{4}$ when $x = 0$, we have from (1)

$$(1 + 1)^3 \times 1 = c \Rightarrow c = 8$$

$\therefore$ The required particular solution is $(1 + e^x)^3 \tan y = 8$.

Example 3. Solve $(x + y + 1)^2 \frac{dy}{dx} = 1$.

Sol. Putting $x + y + 1 = t$, we get $1 + \frac{dy}{dx} = \frac{dt}{dx}$ or $\frac{dy}{dx} = \frac{dt}{dx} - 1$

$\therefore$ The given equation becomes $t^2 \left(\frac{dt}{dx} - 1 \right) = 1$ or $\frac{dt}{dx} = \frac{1+t^2}{t^2}$

$$\Rightarrow \frac{t^2}{1+t^2} dt = dx$$

Integrating, we have $\int \left(1 - \frac{1}{1+t^2} \right) dt = \int dx + c$ or $t - \tan^{-1} t = x + c$

$$\text{or } (x + y + 1) - \tan^{-1}(x + y + 1) = x + c$$

$$\text{or } y = \tan^{-1}(x + y + 1) + C, \text{ where } C = c - 1.$$

TEST YOUR KNOWLEDGE

Solve the following differential equations :

1. $\frac{dy}{dx} = e^{2x+3y}$

2. $xy \frac{dy}{dx} = 1 + x + y + xy$

3. $(x + y)(dx - dy) = dx + dy$

4. $x \frac{dy}{dx} + \cot y = 0$ if $y = \frac{\pi}{4}$ when $x = \sqrt{2}$

5. $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$

6. $y\sqrt{1-x^2} dy + x\sqrt{1-y^2} dx = 0$

7. $\frac{y}{x} \cdot \frac{dy}{dx} = \sqrt{1+x^2+y^2+x^2y^2}$

8. $e^y(1+x^2) \frac{dy}{dx} - 2x(1+e^y) = 0$

9. $\sec^2 x \tan y \, dx + \sec^2 y \tan x \, dy = 0$

10. $(x^2 - yx^2) \frac{dy}{dx} + y^2 + xy^2 = 0$

11. $(1 + x^3) dy - x^2 y dx = 0$, if $y = 2$ when $x = 1$ 12. $a (x dy + y dx) = xy dy$
 13. $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$ 14. $\frac{dy}{dx} = (4x + y + 1)^2$
 15. $(x + y)^2 \frac{dy}{dx} = a^2$ 16. $\sin (x + y) dy = dx$
 17. $\frac{dy}{dx} = \cos (x + y)$ 18. $\frac{dy}{dx} = \sin (x + y) + \cos (x + y)$
 19. $\tan y \frac{dy}{dx} = \sin (x + y) + \sin (x - y)$ 20. $\frac{dy}{dx} - x \tan (y - x) = 1.$

Answers

1. $3e^{2x} + 2e^{-3y} = c$ 2. $y = x + \log [x(1 + y)] + c$ 3. $x + y = ce^{x-y}$
 4. $x \sec y = 2$ 5. $y\sqrt{1-x^2} + x\sqrt{1-y^2} = c$ 6. $\sqrt{1-x^2} + \sqrt{1-y^2} = c$
 7. $\sqrt{1+y^2} = \frac{2}{3}(1+x^2)^{3/2} + c$ 8. $1 + e^y = c(1 + x^2)$ 9. $\tan x \tan y = c$
 10. $\log \left(\frac{x}{y} \right) - \frac{1}{x} - \frac{1}{y} = c$ 11. $y^3 = 4(x^3 + 1)$ 12. $y = a \log (xy) + c$
 13. $e^y = e^x + \frac{x^3}{3} + c$ 14. $4x + y + 1 = 2 \tan (2x + c)$ 15. $x + y = a \tan \left(\frac{y-c}{a} \right)$
 16. $\tan (x + y) - \sec (x + y) = y + c$ 17. $x + c = \tan \left(\frac{x+y}{2} \right)$
 18. $\log \left[1 + \tan \left(\frac{x+y}{2} \right) \right] = x + c$ 19. $2 \cos x + \sec y = c$ 20. $\log \sin (y - x) = \frac{x^2}{2} + c.$

11.6. HOMOGENEOUS EQUATIONS

A differential equation of the form $\frac{dy}{dx} = \frac{f_1(x, y)}{f_2(x, y)}$... (1)

is called a homogeneous differential equation if $f_1(x, y)$ and $f_2(x, y)$ are homogeneous functions of the same degree in x and y .

If $f_1(x, y)$ and $f_2(x, y)$ are homogeneous functions of degree r in x and y , then

$$f_1(x, y) = x^r \phi_1 \left(\frac{y}{x} \right) \quad \text{and} \quad f_2(x, y) = x^r \phi_2 \left(\frac{y}{x} \right)$$

$\therefore$ Equation (1) reduces to $\frac{dy}{dx} = \frac{\phi_1 \left(\frac{y}{x} \right)}{\phi_2 \left(\frac{y}{x} \right)} = F \left(\frac{y}{x} \right)$... (2)

Putting $\frac{y}{x} = v$ i.e., $y = vx$ so that $\frac{dy}{dx} = v + x \frac{dv}{dx}$

Equation (2) becomes $v + x \frac{dv}{dx} = F(v)$

Separating the variables, $\frac{dv}{F(v) - v} = \frac{dx}{x}$

Integrating, we get the solution in terms of v and x . Replacing v by $\frac{y}{x}$, we get the required solution.

ILLUSTRATIVE EXAMPLES

Example 1. Solve $x dy - y dx = \sqrt{x^2 + y^2} dx$.

Sol. The given equation can be written as $\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x}$... (1)

The numerator and denominator on R.H.S. of (1) are homogeneous functions of degree one.

Putting $y = vx$, so that $\frac{dv}{dx} = v + x \frac{dv}{dx}$

Equation (1) becomes $v + x \frac{dv}{dx} = v + \sqrt{1 + v^2}$ or $x \frac{dv}{dx} = \sqrt{1 + v^2}$

Separating the variables, $\frac{dv}{\sqrt{1 + v^2}} = \frac{dx}{x}$

Integrating both sides, $\log(v + \sqrt{1 + v^2}) = \log x + \log c$

or $\log(v + \sqrt{1 + v^2}) = \log(cx)$ or $v + \sqrt{1 + v^2} = cx$

or $\frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}} = cx$ [$\because v = \frac{y}{x}$]

or $y + \sqrt{x^2 + y^2} = cx^2$.

Example 2. Solve $\left(x \tan \frac{y}{x} - y \sec^2 \frac{y}{x} \right) dx + x \sec^2 \frac{y}{x} dy = 0$.

Sol. The given equation can be written as $\frac{dy}{dx} = \frac{y \sec^2 \frac{y}{x} - x \tan \frac{y}{x}}{x \sec^2 \frac{y}{x}} = \frac{y}{x} - \frac{\tan \frac{y}{x}}{\sec^2 \frac{y}{x}}$... (1)

Putting $\frac{y}{x} = v$ i.e., $y = vx$ so that $\frac{dy}{dx} = v + x \frac{dv}{dx}$

Equation (1) becomes
$$v + x \frac{dv}{dx} = v - \frac{\tan v}{\sec^2 v}$$

or
$$\frac{\sec^2 v}{\tan v} dv + \frac{dx}{x} = 0$$

Integrating, we get $\log \tan v + \log x = \log c$

or
$$x \tan v = c$$

or
$$x \tan \frac{y}{x} = c \quad \left[\because v = \frac{y}{x} \right]$$

TEST YOUR KNOWLEDGE

Solve the following differential equations :

1. $(x + y) dx + (y - x) dy = 0$
2. $x \frac{dy}{dx} + \frac{y^2}{x} = y$
3. $(x^2 - y^2) dx = 2xy dy$
4. $(y^2 - 2xy) dx = (x^2 - 2xy) dy$
5. $x(x - y) \frac{dy}{dx} = y(x + y)$
6. $(\sqrt{xy} - x) dy + y dx = 0$
7. $(y^2 + 2xy) dx + (2x^2 + 3xy) dy = 0$
8. $x^2 y dx - (x^3 + y^3) dy = 0$
9. $(x^2 + 2y^2) dx - xy dy = 0$, given that $y = 0$ when $x = 1$
10. $x \frac{dy}{dx} = y (\log y - \log x)$
11. $x \frac{dy}{dx} = y + x \cos^2 \frac{y}{x}$
12. $\frac{dy}{dx} = \frac{y}{x} + \sin \frac{y}{x}$
13. $\left(x \cos \frac{y}{x} + y \sin \frac{y}{x} \right) y - \left(y \sin \frac{y}{x} - x \cos \frac{y}{x} \right) x \frac{dy}{dx} = 0$
14. $(1 + e^{x/y}) dx + e^{x/y} \left(1 - \frac{x}{y} \right) dy = 0$
15. $ye^{x/y} dx = (xe^{x/y} + y^2) dy$
16. $xy \log \left(\frac{x}{y} \right) dx + \left[y^2 - x^2 \log \left(\frac{x}{y} \right) \right] dy = 0.$

Answers

1. $\log(x^2 + y^2) = 2 \tan^{-1} \frac{y}{x} + c$
2. $cx = e^{x/y}$
3. $x(x^2 - 3y^2) = c$
4. $xy(x - y) = c$
5. $\frac{x}{y} + \log \frac{x}{y} = c$
6. $2 \sqrt{\frac{x}{y}} + \log y = c$
7. $xy^2(x + y) = c$
8. $y = ce^{\frac{r^2}{3y^3}}$
9. $x^2 + y^2 = cx^4$
10. $y = xe^{1+cx}$
11. $\tan \frac{y}{x} = \log(cx)$
12. $y = 2x \tan^{-1}(cx)$
13. $xy \cos \frac{y}{x} = c$
14. $x + ye^{x/y} = c$
15. $e^{x/y} = y + c$
16. $\log y - \frac{x^2}{4y^2} \left(2 \log \frac{y}{x} + 1 \right) = c.$

11.7. EQUATIONS REDUCIBLE TO HOMOGENEOUS FORM

A differential equation of the form $\frac{dy}{dx} = \frac{ax + by + c}{a'x + b'y + c'}$... (1)

can be reduced to the homogeneous form as follows :

Case I. When $\frac{a}{a'} \neq \frac{b}{b'}$

Putting $x = X + h, y = Y + k$ (h, k are constants)
so that $dx = dX, dy = dY$

$$\begin{aligned} \text{Equation (1) becomes } \frac{dY}{dX} &= \frac{a(X+h) + b(Y+k) + c}{a'(X+h) + b'(Y+k) + c'} \\ &= \frac{aX + bY + (ah + bk + c)}{a'X + b'Y + (a'h + b'k + c')} \end{aligned} \quad \dots (2)$$

Choose h and k such that (2) becomes homogeneous.

This requires $ah + bk + c = 0$ and $a'h + b'k + c' = 0$

so that
$$\frac{h}{bc' - b'c} = \frac{k}{ca' - c'a} = \frac{1}{ab' - a'b} \quad \text{or} \quad h = \frac{bc' - b'c}{ab' - a'b}, k = \frac{ca' - c'a}{ab' - a'b}$$

Since $\frac{a}{a'} \neq \frac{b}{b'}, \therefore ab' - a'b \neq 0$ so that h, k are finite.

$$\therefore \text{Equation (2) becomes } \frac{dY}{dX} = \frac{aX + bY}{a'X + b'Y}$$

which is homogeneous in X, Y and can be solved by putting $Y = vX$.

Case II. When $\frac{a}{a'} = \frac{b}{b'}, ab' - a'b = 0$ and the above method fails.

Now
$$\frac{a}{a'} = \frac{b}{b'} = \frac{1}{m} \text{ (say) so that } a' = ma, b' = mb$$

$$\text{Equation (1) becomes } \frac{dy}{dx} = \frac{(ax + by) + c}{m(ax + by) + c'} = f(ax + by)$$

which can be solved by putting $ax + by = t$.

ILLUSTRATIVE EXAMPLES

Example 1. Solve $(3y - 7x + 7) dx + (7y - 3x + 3) dy = 0$.

Sol. The given equation can be written as $\frac{dy}{dx} = -\frac{3y - 7x + 7}{7y - 3x + 3} \quad \left[\text{Case } \frac{a}{a'} \neq \frac{b}{b'} \right] \quad \dots (1)$

Putting $x = X + h, y = Y + k$ (h, k are constants)
so that $dx = dX, dy = dY$

Equation (1) becomes
$$\frac{dY}{dX} = -\frac{3(Y+k)-7(X+h)+7}{7(Y+k)-3(X+h)+3}$$

$$= -\frac{3Y-7X+(-7h+3k+7)}{7Y-3X+(-3h+7k+3)} \quad \dots(2)$$

Now, choosing, h, k such that $-7h+3k+7=0$ and $-3h+7k+3=0$
 Solving these equations $h=1, k=0$.

With these values of h, k equation (2) reduces to
$$\frac{dY}{dX} = -\frac{3Y-7X}{7Y-3X} \quad \dots(3)$$

Putting $Y=vX$ so that
$$\frac{dY}{dX} = v + X \frac{dv}{dX}$$

Equation (3) becomes
$$v + X \frac{dv}{dX} = -\frac{3vX-7X}{7vX-3X} \quad \text{or} \quad X \frac{dv}{dX} = \frac{7-3v}{7v-3} - v = \frac{7-7v^2}{7v-3}$$

Separating the variables
$$\frac{7v-3}{1-v^2} dv = 7 \frac{dX}{X} \quad \text{or} \quad \left(\frac{2}{1-v} - \frac{5}{1+v} \right) dv = 7 \frac{dX}{X}$$

Integrating
$$-2 \log(1-v) - 5 \log(1+v) = 7 \log X + c$$

or
$$7 \log X + 2 \log(1-v) + 5 \log(1+v) = -c$$

or
$$\log [X^7 (1-v)^2 (1+v)^5] = -c \quad \text{or} \quad X^7 \left(1 - \frac{Y}{X}\right)^2 \left(1 + \frac{Y}{X}\right)^5 = e^{-c}$$

or
$$(X-Y)^2 (X+Y)^5 = C, \quad \text{where } C = e^{-c} \quad \dots(4)$$

Putting $X = x - h = x - 1, Y = y - k = y$

Equation (4) becomes

$$(x-y-1)^2 (x+y-1)^5 = C$$

Example 2. Solve $(3y+2x+4) dx - (4x+6y+5) dy = 0$.

Sol. The given equation can be written as
$$\frac{dy}{dx} = \frac{(2x+3y)+4}{2(2x+3y)+5} \quad \left[\text{Case } \frac{a}{a'} = \frac{b}{b'} \right] \quad \dots(1)$$

Putting $2x+3y=t$ so that
$$2 + 3 \frac{dy}{dx} = \frac{dt}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{3} \left(\frac{dt}{dx} - 2 \right)$$

Equation (1) becomes
$$\frac{1}{3} \left(\frac{dt}{dx} - 2 \right) = \frac{t+4}{2t+5}$$

or
$$\frac{dt}{dx} = \frac{3t+12}{2t+5} + 2 = \frac{7t+22}{2t+5}$$

Separating the variables
$$\frac{2t+5}{7t+22} dt = dx \quad \text{or} \quad \left(\frac{2}{7} - \frac{9}{7} \cdot \frac{1}{7t+22} \right) dt = dx$$

Integrating both sides
$$\frac{2}{7} t - \frac{9}{49} \log(7t+22) = x + c$$

or
$$14t - 9 \log(7t+22) = 49x + 49c$$

Putting $t = 2x + 3y$, we have $14(2x + 3y) - 9 \log (14x + 21y + 22) = 49x + 49c$

or $7(x - 2y) + 3 \log (14x + 21y + 22) = C. \quad \left(\text{where } C = -\frac{49}{3}c \right)$

TEST YOUR KNOWLEDGE

Solve the following differential equations :

1. $\frac{dy}{dx} = \frac{y + x - 2}{y - x - 4}$
2. $\frac{dy}{dx} = \frac{x + 2y - 3}{2x + y - 3}$
3. $\frac{dy}{dx} + \frac{2x + 3y + 1}{3x + 4y - 1} = 0$
4. $(x + 2y + 3) dx - (2x - y + 1) dy = 0$
5. $(2x - 2y + 5) \frac{dy}{dx} = x - y + 3$
6. $\frac{dy}{dx} = \frac{6x - 4y + 3}{3x - 2y + 1}$
7. $(2x + y + 1) dx + (4x + 2y - 1) dy = 0$
8. $(x + y) (dx - dy) = dx + dy.$

Answers

1. $x^2 + 2xy - y^2 - 4x + 8y = c$
2. $(x - y)^3 = c(x + y - 2)$
3. $x^2 + 3xy + 2y^2 + x - y = c$
4. $\log [(x + 1)^2 + (y + 1)^2] = 4 \tan^{-1} \frac{y + 1}{x + 1} + c$
5. $x - 2y + \log (x - y + 2) = c$
6. $2x - y = \log (3x - 2y + 3) + c$
7. $x + 2y + \log (2x + y - 1) = c$
8. $x - y = \log (x + y) + c.$

11.8. LINEAR DIFFERENTIAL EQUATIONS

A differential equation is said to be linear if the dependent variable and its derivative occur only in the first degree and are not multiplied together.

The general form of a linear differential equation of the first order is $\frac{dy}{dx} + Py = Q \dots (1)$

where P and Q are functions of x only or may be constants.

Equation (1) is also known as *Leibnitz's linear equation*.

To solve it, we multiply both sides by $e^{\int P dx}$, by getting

$$\frac{dy}{dx} e^{\int P dx} + y \left(e^{\int P dx} P \right) = Q e^{\int P dx}$$

or
$$\frac{d}{dx} \left(y e^{\int P dx} \right) = Q e^{\int P dx}$$

Integrating both sides, we have $y e^{\int P dx} = \int Q e^{\int P dx} dx + c$

which is the required solution.

Note 1. The factor $e^{\int P dx}$ on multiplying by which the L.H.S. of (1) becomes the differential co-efficient of a single function is called the integrating factor (briefly written as I.F.) of (1).

Thus I.F. = $e^{\int P dx}$ and the solution is $y \text{ (I.F.)} = \int Q \text{ (I.F.) } dx + c$.

Note 2. Sometimes a differential equation takes linear form if we regard x as dependent variable and y as independent variable. The equation can then be put as $\frac{dx}{dy} + Px = Q$, where P, Q are functions of y only or constants. The integrating factor in this case is $e^{\int P dy}$ and the solution is $x \text{ (I.F.)} = \int Q \text{ (I.F.) } dy + c$.

ILLUSTRATIVE EXAMPLES

Example 1. Solve $x(1-x^2) \frac{dy}{dx} + (2x^2-1)y = x^3$.

Sol. Dividing by $x(1-x^2)$ to make the co-efficient of $\frac{dy}{dx}$ unity, the given equation becomes

$$\frac{dy}{dx} + \frac{2x^2-1}{x(1-x^2)}y = \frac{x^2}{1-x^2}$$

Comparing it with $\frac{dy}{dx} + Py = Q$, we have $P = \frac{2x^2-1}{x(1-x^2)}$, $Q = \frac{x^2}{1-x^2}$

Now
$$P = \frac{2x^2-1}{x(1-x)(1+x)} = -\frac{1}{x} + \frac{1}{2(1-x)} + \frac{1}{2(1+x)}$$

$$\begin{aligned} \int P dx &= -\log x - \frac{1}{2} \log(1-x) - \frac{1}{2} \log(1+x) = -\log [x(1-x)^{1/2}(1+x)^{1/2}] \\ &= -\log [x(1-x^2)^{1/2}] = \log \frac{1}{x\sqrt{1-x^2}} \end{aligned}$$

$$\text{I.F.} = e^{\int P dx} = e^{\log \frac{1}{x\sqrt{1-x^2}}} = \frac{1}{x\sqrt{1-x^2}}$$

Thus the solution is $y \text{ (I.F.)} = \int Q \text{ (I.F.) } dx + c$

or
$$y \cdot \frac{1}{x\sqrt{1-x^2}} = \int \frac{x^2}{1-x^2} \times \frac{1}{x\sqrt{1-x^2}} dx + c$$

$$= -\frac{1}{2} \int (1-x^2)^{-3/2} (-2x) dx + c = (1-x^2)^{-1/2} + c$$

or
$$y = x + cx \sqrt{1-x^2}.$$

Example 2. Solve $(1 + y^2) dx = (\tan^{-1} y - x) dy$.

Sol. The given equation can be written as

$$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1} y}{1+y^2} \text{ which is of the form } \frac{dx}{dy} + Px = Q$$

Here $P = \frac{1}{1+y^2}, Q = \frac{\tan^{-1} y}{1+y^2}$

$$\text{I.F.} = e^{\int P dy} = e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1} y}$$

$$\therefore \text{ The solution is } x (\text{I.F.}) = \int Q (\text{I.F.}) dy + c$$

or
$$xe^{\tan^{-1} y} = \int \frac{\tan^{-1} y}{1+y^2} e^{\tan^{-1} y} dy + c = \int te^t dt + c, \text{ where } t = \tan^{-1} y$$

$$= te^t - \int 1 \cdot e^t dt + c = te^t - e^t + c = (\tan^{-1} y - 1) e^{\tan^{-1} y} + c$$

or
$$x = \tan^{-1} y - 1 + ce^{-\tan^{-1} y}.$$

TEST YOUR KNOWLEDGE

Solve the following differential equations :

- $\frac{dy}{dx} + \frac{y}{x} = x^3 - 3$
- $x \log x \frac{dy}{dx} + y = 2 \log x$
- $(x+1) \frac{dy}{dx} - y = e^x (x+1)^2$
- $(x^2+1) \frac{dy}{dx} + 2xy = x^2$
- $\cos^2 x \frac{dy}{dx} + y = \tan x$
- $(1+x^3) \frac{dy}{dx} + 6x^2y = 1+x^2$
- $\frac{dy}{dx} + y \cot x = \cos x$
- $(1+x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$
- $\frac{dy}{dx} + y \cot x = 4x \operatorname{cosec} x$, if $y\left(\frac{\pi}{2}\right) = 0$
- $\frac{dy}{dx} + y \cot x = 5e^{\cos x}$, if $y\left(\frac{\pi}{2}\right) = -4$
- $\frac{dy}{dx} - y \tan x = 3e^{-\sin x}$, if $y(0) = 4$
- $(1-x^2) \frac{dy}{dx} + 2xy = x \sqrt{1-x^2}$
- $x \frac{dy}{dx} + y = e^x - xy$
- $(1+y^2) + (x - e^{-\tan^{-1} y}) \frac{dy}{dx} = 0$
- $e^{-y} \sec^2 y dy = dx + x dy$
- $(x+2y^3) \frac{dy}{dx} = y$
- $ye^y dx = (y^3 + 2xe^y) dy$
- $\frac{dy}{dx} = \frac{y}{2y \log y + y - x}$

$$19. \quad x^2 \frac{dy}{dx} = 3x^2 - 2xy + 1$$

$$20. \quad \sqrt{1-y^2} dx = (\sin^{-1} y - x) dy$$

Answers

$$1. \quad 10xy = 2x^5 - 15x^2 + c$$

$$2. \quad y \log x = (\log x)^2 + c$$

$$3. \quad y = (x+1)(e^x + c)$$

$$4. \quad y(x^2 + 1) = \frac{x^3}{3} + c$$

$$5. \quad y = \tan x - 1 + ce^{-\tan x}$$

$$6. \quad y(1+x^3)^2 = x + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^6}{6} + c$$

$$7. \quad y \sin x = \frac{1}{2} \sin^2 x + c$$

$$8. \quad 2ye^{\tan^{-1} x} = e^{2 \tan^{-1} x} + c$$

$$9. \quad y \sin x = 2x^2 - \frac{\pi^2}{2}$$

$$10. \quad y \sin x = 1 - 5e^{\cos x}$$

$$11. \quad y \cos x = 7 - 3e^{-\sin x}$$

$$12. \quad y = \sqrt{1-x^2} + c(1-x^2)$$

$$13. \quad xy = \frac{1}{2} e^x + ce^{-x}$$

$$14. \quad x = e^{-\tan^{-1} y} (\tan^{-1} y + c)$$

$$15. \quad xe^y = \tan y + c$$

$$16. \quad x = y^3 + cy$$

$$17. \quad x = y^2 (c - e^{-y})$$

$$18. \quad x = \frac{c}{y} + y \log y.$$

$$19. \quad y = x + x^{-1} + cx^{-2}$$

$$20. \quad x = \sin^{-1} y - 1 + ce^{-\sin^{-1} y}.$$

11.9. EQUATIONS REDUCIBLE TO THE LINEAR FORM (BERNOULLI'S EQUATION)

$$(i) \text{ An equation of the form } \frac{dy}{dx} + Py = Qy^n \quad \dots(1)$$

where P and Q are functions of x only or constants is known as *Bernoulli's equation*. Though not linear, it can be made linear.

$$\text{Dividing both sides of (1) by } y^n, \text{ we have } y^{-n} \frac{dy}{dx} + Py^{1-n} = Q \quad \dots(2)$$

$$\text{Putting } y^{1-n} = z \text{ so that } (1-n)y^{-n} \frac{dy}{dx} = \frac{dz}{dx}$$

$$\text{or } y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dz}{dx}$$

$$\text{Equation (2) becomes } \frac{1}{1-n} \frac{dz}{dx} + Pz = Q \quad \text{or} \quad \frac{dz}{dx} + (1-n)Pz = (1-n)Q$$

which is a linear differential equation with z as the dependent variable.

$$(ii) \text{ General equation reducible to linear form is } f'(y) \frac{dy}{dx} + P f(y) = Q \quad \dots(1)$$

where P and Q are functions of x only or constants.

$$\text{Putting } f(y) = z \text{ so that } f'(y) \frac{dy}{dx} = \frac{dz}{dx}$$

$$\text{Equation (1) becomes } \frac{dz}{dx} + Pz = Q, \quad \text{which is linear.}$$

ILLUSTRATIVE EXAMPLES

Example 1. Solve $\frac{dy}{dx} + \frac{x}{1-x^2} y = x\sqrt{y}$.

Sol. Dividing by $\sqrt{y}$, the given equation becomes $y^{-1/2} \frac{dy}{dx} + \frac{x}{1-x^2} y^{1/2} = x$... (1)

Putting $y^{1/2} = z$ so that $\frac{1}{2} y^{-1/2} \frac{dy}{dx} = \frac{dz}{dx}$ or $y^{-1/2} \frac{dy}{dx} = 2 \frac{dz}{dx}$

Equation (1) becomes $2 \frac{dz}{dx} + \frac{x}{1-x^2} z = x$

or $\frac{dz}{dx} + \frac{x}{2(1-x^2)} z = \frac{x}{2}$ which is linear in z .

$$\text{I.F.} = e^{\int \frac{x}{2(1-x^2)} dx} = e^{-\frac{1}{4} \int \frac{-2x}{1-x^2} dx} = e^{-\frac{1}{4} \log(1-x^2)} = (1-x^2)^{-1/4}$$

$\therefore$ The solution is $z (\text{I.F.}) = \int \frac{x}{2} (\text{I.F.}) dx + c$

$$\begin{aligned} \text{or } z(1-x^2)^{-1/4} &= \int \frac{x}{2} (1-x^2)^{-1/4} dx + c \\ &= -\frac{1}{4} \int (1-x^2)^{-1/4} (-2x) dx + c = -\frac{1}{4} \cdot \frac{(1-x^2)^{3/4}}{3/4} + c \end{aligned}$$

$$\text{or } z = -\frac{1}{3} (1-x^2) + c (1-x^2)^{1/4}$$

$$\text{or } \sqrt{y} = -\frac{1}{3} (1-x^2) + c (1-x^2)^{1/4} \quad \left[\because z = \sqrt{y} \right]$$

Example 2. Solve $xy(1+xy^2) \frac{dy}{dx} = 1$.

Sol. The given equation can be written as $\frac{dx}{dy} - yx = y^3 x^2$

Dividing by x^2 , we have $x^{-2} \frac{dx}{dy} - yx^{-1} = y^3$... (1)

Putting $x^{-1} = z$ so that $x^{-2} \frac{dx}{dy} = -\frac{dz}{dy}$

Equation (1) becomes $-\frac{dz}{dy} - yz = y^3$

or $\frac{dz}{dy} + yz = -y^3$ which is linear in z .

$$\text{I.F.} = e^{\int y \, dy} = e^{\frac{1}{2}y^2}$$

∴ The solution is $z (\text{I.F.}) = \int -y^3 (\text{I.F.}) \, dy + c$

$$\begin{aligned} \text{or } z \cdot e^{\frac{1}{2}y^2} &= \int -y^3 e^{\frac{1}{2}y^2} \, dy + c \\ &= - \int y^2 e^{\frac{1}{2}y^2} \cdot y \, dy + c = - \int 2t e^t \, dt + c \quad \text{where } t = \frac{1}{2}y^2 \\ &= -2 (te^t - e^t) + c = -2e^{\frac{1}{2}y^2} \left(\frac{1}{2}y^2 - 1 \right) + c \end{aligned}$$

$$\text{or } z = -2 \left(\frac{1}{2}y^2 - 1 \right) + c e^{-\frac{1}{2}y^2}$$

$$\text{or } \frac{1}{x} = 2 - y^2 + c e^{-\frac{1}{2}y^2}.$$

Example 3. Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$.

Sol. Dividing by $\cos^2 y$, we have $\sec^2 y \frac{dy}{dx} + x \frac{2 \sin y \cos y}{\cos^2 y} = x^3$

$$\text{or } \sec^2 y \frac{dy}{dx} + 2x \tan y = x^3 \quad \dots(1)$$

Putting $\tan y = z$ so that $\sec^2 y \frac{dy}{dx} = \frac{dz}{dx}$

Equation (1) becomes $\frac{dz}{dx} + 2xz = x^3$ which is linear in z .

$$\text{I.F.} = e^{\int 2x \, dx} = e^{x^2}$$

∴ The solution is

$$\begin{aligned} z \cdot e^{x^2} &= \int x^3 \cdot e^{x^2} \, dx + c = \int x^2 e^{x^2} \cdot x \, dx + c \\ &= \frac{1}{2} \int t e^t \, dt + c \quad \text{where } t = x^2 \\ &= \frac{1}{2} (t - 1) e^t + c = \frac{1}{2} (x^2 - 1) e^{x^2} + c \end{aligned}$$

$$\text{or } z = \frac{1}{2} (x^2 - 1) + c e^{-x^2}$$

$$\text{or } \tan y = \frac{1}{2} (x^2 - 1) + c e^{-x^2} \quad (\because z = \tan y)$$

TEST YOUR KNOWLEDGE

Solve the following differential equations :

1. $x \frac{dy}{dx} + y = x^3 y^6$
2. $2 \frac{dy}{dx} = \frac{y}{x} + \frac{y^2}{x^2}$
3. $\frac{dy}{dx} - x^2 y = y^2 e^{-\frac{1}{3}x^3}$
4. $(x^3 y^2 + xy) dx = dy$
5. (a) $e^y \left(\frac{dy}{dx} + 1 \right) = e^x$
- (b) $(x+1) \frac{dy}{dx} + 1 = 2e^{-y}$
6. $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x) e^x \sec y$
7. $(xy^2 - e^{x^{\frac{1}{3}}}) dx - x^2 y dy = 0$
8. $\frac{dy}{dx} + y \tan x = y^3 \cos x$
9. $\frac{dy}{dx} + \frac{y \log y}{x} = \frac{y (\log y)^2}{x^2}$
10. $y - \cos x \frac{dy}{dx} = y^2 (1 - \sin x) \cos x$, given that $y = 2$ when $x = 0$
11. $y (2xy + e^x) dx = e^x dy$
12. $\tan y \frac{dy}{dx} + \tan x = \cos y \cos^2 x$.

Answers

1. $\frac{1}{y^5} = \frac{5}{2} x^3 + cx^5$
2. $\frac{x}{y} = 1 + c\sqrt{x}$
3. $y(c-x) = e^{\frac{1}{3}x^3}$
4. $\frac{1}{y} = -x^2 + 2 + ce^{-\frac{1}{2}x^2}$
5. (a) $e^{x+y} = \frac{1}{2} e^{2x} + c$
- (b) $(x+1) e^y = 2x + c$
6. $\sin y = (1+x)(e^x + c)$
7. $3y^2 = x^2 (2e^{x^{\frac{1}{3}}} + c)$
8. $\cos^2 x = y^2 (c - 2 \sin x + \frac{2}{3} \sin^3 x)$
9. $\frac{1}{x \log y} = \frac{1}{2x^2} + c$
10. $2 (\tan x + \sec x) = y(2 \sin x + 1)$
11. $e^x = y(c - x^2)$
12. $\sec y = (c + \sin x) \cos x$.

11.10. EXACT DIFFERENTIAL EQUATIONS

A differential equation obtained from its primitive directly by differentiation, without any operation of multiplication, elimination or reduction etc. is said to be an exact differential equation.

Thus a differential equation of the form $M(x, y) dx + N(x, y) dy = 0$ is an exact differential equation if it can be obtained directly by differentiating the equation $u(x, y) = c$, which is its primitive.

i.e., if $du = Mdx + Ndy$

11.11. THEOREM

The necessary and sufficient condition for the differential equation $Mdx + Ndy = 0$ to be exact is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

The condition is necessary.

The equation $Mdx + Ndy = 0$ will be exact, if $du \equiv Mdx + Ndy$

$$\text{But} \quad du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$\therefore \quad Mdx + Ndy = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

Equating co-efficients of dx and dy , we get

$$M = \frac{\partial u}{\partial x} \quad \text{and} \quad N = \frac{\partial u}{\partial y}$$

$$\therefore \quad \frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial y \partial x} \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x \partial y}$$

$$\text{But} \quad \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$$

$$\therefore \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

which is the necessary condition of exactness.

The condition is sufficient.

$$\text{Let} \quad u = \int_{y \text{ constant}} M dx$$

$$\therefore \quad \frac{\partial u}{\partial x} = M \quad \text{and} \quad \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial M}{\partial y}$$

$$\text{But} \quad \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y} \quad \text{and} \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\therefore \quad \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right)$$

Integrating both sides w.r.t. x , treating y as constant, we have $N = \frac{\partial u}{\partial y} + f(y)$

$$\begin{aligned} \therefore \quad Mdx + Ndy &= \frac{\partial u}{\partial x} dx + \left\{ \frac{\partial u}{\partial y} + f(y) \right\} dy && \left[\because M = \frac{\partial u}{\partial x}, N = \frac{\partial u}{\partial y} + f(y) \right] \\ &= \left(\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \right) + f(y) dy = du + f(y) dy = d[u + \int f(y) dy] \end{aligned}$$

which shows that $Mdx + Ndy$ is an exact differential and hence $Mdx + Ndy = 0$ is an exact differential equation.

Note. Since $Mdx + Ndy = d[u + \int f(y) dy]$

$$\therefore \quad Mdx + Ndy = 0 \Rightarrow d[u + \int f(y) dy] = 0$$

Integrating $u + \int f(y) dy = c$

$$\text{But} \quad u = \int_{y \text{ constant}} M dx \quad \text{and} \quad f(y) = \text{terms of } N \text{ not containing } x$$

Hence the solution of $Mdx + Ndy = 0$ is

$$\int_{y \text{ constant}} Mdx + \int (\text{terms of } N \text{ not containing } x) dy = c.$$

ILLUSTRATIVE EXAMPLES

Example 1. Solve $(5x^4 + 3x^2y^2 - 2xy^3) dx + (2x^3y - 3x^2y^2 - 5y^4) dy = 0$.

Sol. Here $M = 5x^4 + 3x^2y^2 - 2xy^3$ and $N = 2x^3y - 3x^2y^2 - 5y^4$

$$\therefore \frac{\partial M}{\partial y} = 6x^2y - 6xy^2 = \frac{\partial N}{\partial x}$$

Thus the given equation is exact and its solution is

$$\int_{y \text{ constant}} Mdx + \int (\text{terms of } N \text{ not containing } x) dy = c$$

$$i.e., \int_{y \text{ constant}} (5x^4 + 3x^2y^2 - 2xy^3) dx + \int -5y^4 dy = c$$

$$or \quad x^5 + x^3y^2 - x^2y^3 - y^5 = c.$$

Example 2. Solve $[\cos x \tan y + \cos (x + y)] dx + [\sin x \sec^2 y + \cos (x + y)] dy = 0$.

Sol. Here $M = \cos x \tan y + \cos (x + y)$

and $N = \sin x \sec^2 y + \cos (x + y)$

$$\therefore \frac{\partial M}{\partial y} = \cos x \sec^2 y - \sin (x + y) = \frac{\partial N}{\partial x}$$

Thus the given equation is exact and its solution is

$$\int_{y \text{ constant}} Mdx + \int (\text{terms of } N \text{ not containing } x) dy = c$$

$$i.e., \int_{y \text{ constant}} [\cos x \tan y + \cos (x + y)] dx = c$$

$$or \quad \sin x \tan y + \sin (x + y) = c.$$

TEST YOUR KNOWLEDGE

Solve the following differential equations :

- $(x^2 - 4xy - 2y^2) dx + (y^2 - 4xy - 2x^2) dy = 0$
- $(3x^2 + 6xy^2) dx + (6x^2y + 4y^3) dy = 0$
- $(x^2 + y^2 - a^2) xdx + (x^2 - y^2 - b^2) ydy = 0$
- $(1 + e^{x/y} dx + \left(1 - \frac{x}{y}\right) e^{x/y} dy = 0$
- $(y \cos x + 1) dx + \sin x dy = 0$
- $(\sec x \tan x \tan y - e^x) dx + \sec x \sec^2 y dy = 0$
- $ye^{xy} dx + (xe^{xy} + 2y) dy = 0$
- $(y^2e^{xy^2} + 4x^3) dx + (2xye^{xy^2} - 3y^2) dy = 0$
- $(2xy \cos x^2 - 2xy + 1) dx + (\sin x^2 - x^2) dy = 0$
- $(\sin x \cos y + e^{2x}) dx + (\cos x \sin y + \tan y) dy = 0$

$$11. \left[y \left(1 + \frac{1}{x} \right) + \cos y \right] dx + (x + \log x - x \sin y) dy = 0 \quad 12. \frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$$

$$13. x dy + y dx + \frac{xdy - ydx}{x^2 + y^2} = 0.$$

Answers

$$\begin{array}{lll} 1. x^3 - 6x^2y - 6xy^2 + y^3 = c & 2. x^3 + 3x^2y^2 + y^4 = c & 5. y \sin x + x = c \\ 3. x^4 + 2x^2y^2 - 2a^2x^2 - y^4 - 2b^2y^2 = c & 4. x + ye^{x/y} = c & 6. \sec x \tan y - e^x = c \\ 7. e^{xy} + y^2 = c & 8. e^{xy^2} + x^4 - y^3 = c & \\ 9. y \sin x^2 - x^2y + x = c & 10. -\cos x \cos y + \frac{1}{2}e^{2x} + \log \sec y = c & \\ 11. y(x + \log x) + x \cos y = c & 12. y \sin x + (\sin y + y)x = c & 13. xy + \tan^{-1} \frac{y}{x} = c. \end{array}$$

11.12. EQUATIONS REDUCIBLE TO EXACT EQUATIONS

Differential equations which are not exact can sometimes be made exact after multiplying by a suitable factor (a function of x and/or y) called the integrating factor.

For example, consider the equation $y dx - x dy = 0$... (1)

Here $M = y$ and $N = -x$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}, \text{ therefore the equation is not exact.}$$

$$(i) \text{ Multiplying the equation by } \frac{1}{y^2}, \text{ it becomes } \frac{ydx - xdy}{y^2} = 0 \quad \text{or} \quad d\left(\frac{x}{y}\right) = 0$$

which is exact.

$$(ii) \text{ Multiplying the equation by } \frac{1}{x^2}, \text{ it becomes } \frac{ydx - xdy}{x^2} = 0 \quad \text{or} \quad d\left(\frac{y}{x}\right) = 0$$

which is exact.

$$(iii) \text{ Multiplying the equation by } \frac{1}{xy}, \text{ it becomes } \frac{dx}{x} - \frac{dy}{y} = 0 \quad \text{or} \quad d(\log x - \log y) = 0$$

which is exact.

$$\therefore \frac{1}{y^2}, \frac{1}{x^2} \text{ and } \frac{1}{xy} \text{ are integrating factors of (1).}$$

If a differential equation has one integrating factor, it has an infinite number of integrating factors.

(a) **I.F. found by inspection.** In a number of problems, a little analysis helps to find the integrating factor. The following differentials are useful in selecting a suitable integrating factor.

$$(i) ydx + xdy = d(xy)$$

$$(ii) \frac{xdy - ydx}{x^2} = d\left(\frac{y}{x}\right)$$

$$(iii) \frac{ydx - xdy}{y^2} = d\left(\frac{x}{y}\right)$$

$$(iv) \frac{xdy - ydx}{x^2 + y^2} = d\left(\tan^{-1} \frac{y}{x}\right)$$

$$(v) \frac{xdy - ydx}{xy} = d \left[\log \left(\frac{y}{x} \right) \right]$$

$$(vi) \frac{ydx + xdy}{xy} = d [\log (xy)]$$

$$(vii) \frac{xdy + ydy}{x^2 + y^2} = d \left[\frac{1}{2} \log (x^2 + y^2) \right]$$

$$(viii) \frac{xdy - ydx}{x^2 - y^2} = d \left(\frac{1}{2} \log \frac{x+y}{x-y} \right).$$

ILLUSTRATIVE EXAMPLES

Example 1. Solve $ydx - xdy + 3x^2y^2e^{x^3} dx = 0$.

Sol. Since $3x^2e^{x^3} = d(e^{x^3})$, the term $3x^2y^2e^{x^3} dx$ should not involve y^2 .

This suggests that $\frac{1}{y^2}$ may be an I.F.

Multiplying throughout by $\frac{1}{y^2}$, we have $\frac{ydx - xdy}{y^2} + 3x^2e^{x^3} dx = 0$

$$d \left(\frac{x}{y} \right) + d(e^{x^3}) = 0, \text{ which is exact.}$$

Integrating, we get $\frac{x}{y} + e^{x^3} = c$.

Example 2. Solve $xdy - ydx = x\sqrt{x^2 - y^2} dx$.

Sol. The given equation is $xdy - ydx = x^2 \sqrt{1 - \left(\frac{y}{x}\right)^2} dx$

$$\text{or } \frac{\frac{xdy - ydx}{x^2}}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} = dx$$

$$\text{or } d \left(\sin^{-1} \frac{y}{x} \right) = dx, \text{ which is exact.}$$

Integrating, we get $\sin^{-1} \frac{y}{x} = x + c$ or $y = x \sin (x + c)$.

(b) **I.F. for a homogeneous equation.** If $Mdx + Ndy = 0$ is a homogeneous equation in x and y , then $\frac{1}{Mx + Ny}$ is an I.F. provided $Mx + Ny \neq 0$.

Note. If $Mx + Ny$ consists of only one term, use the above method of I.F. otherwise, proceed by putting $y = vx$.

Example. Solve $(x^2y - 2xy^2) dx - (x^3 - 3x^2y) dy = 0$.

Sol. The given equation is homogeneous in x and y with

$$M = x^2y - 2xy^2 \text{ and } N = -x^3 + 3x^2y$$

Now $Mx + Ny = x^3y - 2x^2y^2 - x^2y + 3x^2y^2 = x^2y^2 \neq 0$

$$\therefore \text{I.F.} = \frac{1}{Mx + Ny} = \frac{1}{x^2 y^2}$$

Multiplying throughout by $\frac{1}{x^2 y^2}$, the given equation becomes

$$\left(\frac{1}{y} - \frac{2}{x}\right) dx - \left(\frac{x}{y^2} - \frac{3}{y}\right) dy = 0$$

which is exact. The solution is $\int_{y \text{ constant}} \left(\frac{1}{y} - \frac{2}{x}\right) dx + \int \frac{3}{y} dy = c$

or
$$\frac{x}{y} - 2 \log x + 3 \log y = c.$$

(c) **I.F. for an equation of the form $f_1(xy) y dx + f_2(xy) x dy = 0$.**

If $Mdx + Ndy = 0$ is of the form $f_1(xy) y dx + f_2(xy) x dy = 0$ then $\frac{1}{Mx - Ny}$ is an I.F. provided $Mx - Ny \neq 0$.

Example. Solve $y(xy + 2x^2y^2) dx + x(xy - x^2y^2) dy = 0$.

Sol. The given equation is of the form $f_1(xy) y dx + f_2(xy) x dy = 0$.

Here $M = xy^2 + 2x^2y^2$ and $N = x^2y - x^3y^2$

Now $Mx - Ny = x^2y^2 + 2x^3y^3 - x^2y^2 + x^3y^3 = 3x^3y^3 \neq 0$

$$\therefore \text{I.F.} = \frac{1}{Mx - Ny} = \frac{1}{3x^3y^3}$$

Multiplying throughout by $\frac{1}{3x^3y^3}$, the given equation becomes

$$\left(\frac{1}{3x^2y} + \frac{2}{3x}\right) dx + \left(\frac{1}{3xy^2} - \frac{1}{3y}\right) dy = 0$$

which is exact. The solution is $\int_{y \text{ constant}} \left(\frac{1}{3x^2y} + \frac{2}{3x}\right) dx + \int -\frac{1}{3y} dy = c$

or
$$-\frac{1}{3xy} + \frac{2}{3} \log x - \frac{1}{3} \log y = c$$

or
$$-\frac{1}{xy} + 2 \log x - \log y = C, \text{ where } C = 3c.$$

(d) **For the equation $Mdx + Ndy = 0$.**

(i) If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$ is a function of x only, say $f(x)$, then $e^{\int f(x) dx}$ is an I.F.

(ii) If $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$ is a function of y only, we say $g(y)$, then $e^{\int g(y) dy}$ is an I.F.

ILLUSTRATIVE EXAMPLES

Example 1. Solve $(xy^2 - e^{x^{\frac{1}{3}}}) dx - x^2y dy = 0$.

Sol. Here $M = xy^2 - e^{x^{\frac{1}{3}}}$ and $N = -x^2y$

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2xy - (-2xy)}{-x^2y} = -\frac{4}{x}$$

which is a function of x only.

$$\therefore \text{I.F.} = e^{\int -\frac{4}{x} dx} = e^{-4 \log x} = \frac{1}{x^4}$$

Multiplying throughout by $\frac{1}{x^4}$, we have $\left(\frac{y^2}{x^3} - \frac{1}{x^4} e^{\frac{1}{x^3}} \right) dx - \frac{y}{x^2} dy = 0$

which is exact. The solution is

$$\int_{y \text{ constant}} \left(\frac{y^2}{x^3} - \frac{1}{x^4} e^{\frac{1}{x^3}} \right) dx = c$$

or $-\frac{y^2}{2x^2} + \frac{1}{3} \int -\frac{3}{x^4} e^{\frac{1}{x^3}} dx = c$ or $-\frac{y^2}{2x^2} + \frac{1}{3} \int e^t dt = c$, where $t = \frac{1}{x^3}$

or $-\frac{y^2}{2x^2} + \frac{1}{3} e^{1/x^3} = c.$

Example 2. Solve $(xy^3 + y) dx + 2(x^2y^2 + x + y^4) dy = 0$.

Sol. Here $M = xy^3 + y$ and $N = 2x^2y^2 + 2x + 2y^4$

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{4xy^2 + 2 - 3xy^2 - 1}{xy^3 + y} = \frac{xy^2 + 1}{y(xy^2 + 1)} = \frac{1}{y}$$

which is a function of y only.

$$\therefore \text{I.F.} = e^{\int \frac{1}{y} dy} = e^{\log y} = y$$

Multiplying throughout by y , we have $(xy^4 + y^2) dx + 2(x^2y^3 + xy + y^5) dy = 0$

which is exact. The solution is $\int_{y \text{ constant}} (xy^4 + y^2) dx + \int 2y^5 dy = c$

or $\frac{x^2y^4}{2} + xy^2 + \frac{y^6}{3} = c.$

(e) **I.F. for the equation of the form $x^a y^b (my dx + nx dy) + x^c y^d (py dx + qx dy) = 0$**

where a, b, c, d, m, n, p, q are all constant is $x^h y^k$, where h, k are so chosen that after multiplication by $x^h y^k$ the equation becomes exact.

Example. Solve $(2x^2 y^2 + y) dx + (3x - x^3 y) dy = 0$.

Sol. The equation can be written as $2(x^2 y^2 dx - x^3 y dy) + (y dx + 3x dy) = 0$

or $x^2 y (2y dx - x dy) + x^0 y^0 (y dx + 3x dy) = 0$

which is of the form mentioned above. Therefore, it has an I.F. of the form $x^h y^k$.

Multiplying the given equation by $x^h y^k$, we have

$$(2x^{h+2} y^{k+2} + x^h y^{k+1}) dx + (3x^{h+1} y^k - x^{h+3} y^{k+1}) dy = 0$$

For this equation to be exact, we must have

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ i.e.,}$$

$$2(k+2)x^{h+2}y^{k+1} + (k+1)x^h y^k = 3(h+1)x^h y^k - (h+3)x^{h+2}y^{k+1}$$

which holds when $2(k+2) = -(h+3)$ and $k+1 = 3(h+1)$

i.e., when $h + 2k + 7 = 0$ and $3h - k + 2 = 0$

Solving these equation, we have $h = -\frac{11}{7}, k = -\frac{19}{7}$

$$\therefore \text{I.F.} = x^{-\frac{11}{7}} y^{-\frac{19}{7}}$$

Multiplying the given equation by $x^{-\frac{11}{7}} y^{-\frac{19}{7}}$, we have

$$\left(2x^{\frac{3}{7}} y^{-\frac{5}{7}} + x^{-\frac{11}{7}} y^{-\frac{12}{7}} \right) dx + \left(3x^{-\frac{4}{7}} y^{-\frac{19}{7}} - x^{\frac{10}{7}} y^{-\frac{12}{7}} \right) dy = 0$$

which is exact. The solution is $\int_{y \text{ constant}} \left(2x^{\frac{3}{7}} y^{-\frac{5}{7}} + x^{-\frac{11}{7}} y^{-\frac{12}{7}} \right) dx = c$

$$\text{or } \frac{7}{5} x^{\frac{10}{7}} y^{-\frac{5}{7}} - \frac{7}{4} x^{-\frac{4}{7}} y^{-\frac{12}{7}} = c \quad \text{or} \quad 4x^{\frac{10}{7}} y^{-\frac{5}{7}} - 5x^{-\frac{4}{7}} y^{-\frac{12}{7}} = C \quad \text{where } C = \frac{20}{7}c.$$

Note. The values of h and k can also be determined from the relations

$$\frac{a+h+1}{m} = \frac{b+k+1}{n} \quad \text{and} \quad \frac{c+h+1}{p} = \frac{d+k+1}{q}.$$

TEST YOUR KNOWLEDGE

Solve the following differential equations :

1. $x dy - y dx = (x^2 + y^2) dx$
2. $y (2xy + e^x) dx - e^x dy = 0$
3. $(1 + xy) y dx + (1 - xy) x dy = 0$
4. $x dy - y dx = xy^2 dx$
5. $(xye^{x/y} + y^2) dx - x^2 e^{x/y} dy = 0$
6. $x^2 y dx - (x^3 + y^3) dy = 0$

7. $(3xy^2 - y^3) dx - (2x^2y - xy^2) dy = 0$ 8. $(x^2y^2 + xy + 1) ydx + (x^2y^2 - xy + 1) xdy = 0$
 9. $y(2xy + 1) dx + x(1 + 2xy - x^3y^3) dy = 0$ 10. $(x^2 + y^2 + 2x) dx + 2ydy = 0$
 11. $(x^2 + y^2 + 1) dx - 2xydy = 0$ 12. $\left(y + \frac{y^3}{3} + \frac{x^2}{2}\right) dx + \frac{1}{4}(x + xy^2) dy = 0$
 13. $(y^4 + 2y) dx + (xy^3 + 2y^4 - 4x) dy = 0$ 14. $(3xy - 2ay^2) dx + (x^2 - 2axy) dy = 0$
 15. $ydx - xdy + \log x dx = 0$ 16. $(xy^2 + 2x^2y^3) dx + (x^2y - x^3y^2) dy = 0$
 17. $(2x^2y - 3y^4) dx + (3x^3 + 2xy^3) dy = 0$.

Answers

1. $y = x \tan(x + c)$ 2. $x^2 + \frac{e^x}{y} = c$ 3. $-\frac{1}{xy} + \log \frac{x}{y} = c$ 4. $\frac{x^2}{2} + \frac{x}{y} = c$
 5. $e^{x/y} + \log x = c$ 6. $\log y - \frac{x^3}{3y^3} = c$ 7. $3 \log x - 2 \log y + \frac{y}{x} = c$
 8. $xy + \log \frac{x}{y} - \frac{1}{xy} = c$ 9. $\frac{1}{x^2y^2} + \frac{1}{3x^3y^3} + \log y = c$ 10. $e^x(x^2 + y^2) = c$
 11. $x - \frac{y^2}{x} - \frac{1}{x} = c$ 12. $3x^4y + x^4y^3 + x^6 = c$ 13. $\left(y + \frac{2}{y^2}\right)x + y^2 = c$ 14. $x^2(ay^2 - xy) = c$
 15. $cx + y + \log x + 1 = 0$ 16. $-\frac{1}{xy} + 2 \log x - \log y = c$
 17. $5x^{\frac{36}{13}}y^{\frac{24}{13}} - 12x^{\frac{10}{13}}y^{\frac{15}{13}} = c$.

11.13. DIFFERENTIAL EQUATIONS OF THE FIRST ORDER AND HIGHER DEGREE

So far, we have discussed differential equations of the first order and first degree. Now we shall study differential equations of the first order and degree higher than the first. For convenience, we denote $\frac{dy}{dx}$ by p .

A differential equation of the first order and n^{th} degree is of the form

$$p^n + P_1p^{n-1} + P_2p^{n-2} + \dots + P_n = 0 \quad \dots(1)$$

where $P_1, P_2, \dots, P_n$ are functions of x and y .

Since it is a differential equation of the first order, its general solution will contain only one arbitrary constant.

In the various cases which follow, the problem is reduced to that of solving one or more equations of the first order and first degree.

11.14. EQUATIONS SOLVABLE FOR p

Resolving the left hand side of (1) into n linear factors, we have

$$[p - f_1(x, y)][p - f_2(x, y)], \dots, [p - f_n(x, y)] = 0$$

which is equivalent to $p - f_1(x, y) = 0, p - f_2(x, y) = 0, \dots, p - f_n(x, y) = 0$

Each of these equations is of the first order and first degree and can be solved by the methods already discussed.

If the solutions of the above n component equations are

$$F_1(x, y, c) = 0, F_2(x, y, c) = 0, \dots, F_n(x, y, c) = 0$$

then the general solution of (1) is given by

$$F_1(x, y, c), F_2(x, y, c), \dots, F_n(x, y, c) = 0.$$

ILLUSTRATIVE EXAMPLES

Example 1. Solve $x^2 \left(\frac{dy}{dx} \right)^2 + xy \frac{dy}{dx} - 6y^2 = 0$.

Sol. The given equation is $x^2 p^2 + xyp - 6y^2 = 0$ where $p = \frac{dy}{dx}$

Factorising $(xp + 3y)(xp - 2y) = 0$

$$\Rightarrow xp + 3y = 0 \quad \text{or} \quad xp - 2y = 0$$

Now $xp + 3y = 0$

$$\Rightarrow x \frac{dy}{dx} + 3y = 0 \quad \text{or} \quad \frac{dy}{y} + 3 \frac{dx}{x} = 0$$

Integrating $\log y + 3 \log x = \log c$ or $x^3 y = c$

Also $xp - 2y = 0$

$$\Rightarrow x \frac{dy}{dx} - 2y = 0 \quad \text{or} \quad \frac{dy}{y} - 2 \frac{dx}{x} = 0$$

Integrating $\log y - 2 \log x = \log c$ or $\frac{y}{x^2} = c$ or $y = cx^2$

$\therefore$ The general solution of the given equation is $(x^3 y - c)(y - cx^2) = 0$.

Example 2. Solve $xyp^2 + p(3x^2 - 2y^2) - 6xy = 0$.

Sol. Solving the given equation for p , we have

$$p = \frac{-(3x^2 - 2y^2) \pm \sqrt{(3x^2 - 2y^2)^2 + 24x^2 y^2}}{2xy}$$

$$= \frac{(2y^2 - 3x^2) \pm (3x^2 + 2y^2)}{2xy} = \frac{2y}{x} \quad \text{or} \quad -\frac{3x}{y}$$

Now $p = \frac{2y}{x} \Rightarrow \frac{dy}{dx} = \frac{2y}{x} \quad \text{or} \quad \frac{dy}{y} - \frac{2dx}{x} = 0$

Integrating $\log y - 2 \log x = \log c$ or $\frac{y}{x^2} = c$ or $y = cx^2$

Also $p = -\frac{3x}{y} \Rightarrow \frac{dy}{dx} = -\frac{3x}{y} \quad \text{or} \quad ydy + 3xdx = 0$

Integrating $\frac{y^2}{2} + \frac{3x^2}{2} = C$ or $y^2 + 3x^2 = c$

$\therefore$ The general solution of the given equation is $(y - cx^2)(y^2 + 3x^2 - c) = 0$.

Example 3. Solve $p^2 + 2py \cot x = y^2$.

Sol. The given equation can be written as $(p + y \cot x)^2 = y^2 (1 + \cot^2 x)$

or $p + y \cot x = \pm y \operatorname{cosec} x$

$\therefore$ The component equations are

$$p = y (-\cot x + \operatorname{cosec} x) \quad \dots(1)$$

and $p = y (-\cot x - \operatorname{cosec} x) \quad \dots(2)$

From (1), $\frac{dy}{dx} = y (-\cot x + \operatorname{cosec} x)$

or $\frac{dy}{y} = (-\cot x + \operatorname{cosec} x) dx$

Integrating $\log y = -\log \sin x + \log \tan \frac{x}{2} + \log c = \log \frac{c \tan \frac{x}{2}}{\sin x}$

or $y = \frac{c \tan \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}} = \frac{c}{2 \cos^2 \frac{x}{2}} = \frac{c}{1 + \cos x}$

or $y(1 + \cos x) = c$

From (2), $\frac{dy}{dx} = y(-\cot x - \operatorname{cosec} x)$

or $\frac{dy}{y} = (-\cot x - \operatorname{cosec} x) dx$

Integrating $\log y = -\log \sin x - \log \tan \frac{x}{2} + \log c = \log \frac{c}{\sin x \tan \frac{x}{2}}$

or $y = \frac{c}{2 \sin^2 \frac{x}{2}} = \frac{c}{1 - \cos x}$ or $y(1 - \cos x) = c$

$\therefore$ The general solution of the given equation is $y(1 \pm \cos x) = c$.

TEST YOUR KNOWLEDGE

Solve the following equations :

1. $p^2 - 7p + 12 = 0$

2. $xy \left(\frac{dy}{dx} \right)^2 - (x^2 + y^2) \frac{dy}{dx} + xy = 0$

3. $yp^2 + (x - y)p - x = 0$

4. $x^2 \left(\frac{dy}{dx} \right)^2 + 3xy \frac{dy}{dx} + 2y^2 = 0$

5. $\frac{dy}{dx} - \frac{dx}{dy} = \frac{x}{y} - \frac{y}{x}$

6. $p^2 - 2p \sinh x - 1 = 0$

7. $p(p + y) = x(x + y)$

8. $4y^2p^2 + 2pxy(3x + 1)3x^3 = 0$.

Answers

1. $(y - 4x - c)(y - 3x - c) = c$ 2. $(y^2 - x^2 - c)(y - cx) = 0$ 3. $(y - x - c)(x^2 + y^2 - c) = 0$
 4. $(xy - c)(x^2y - c) = 0$ 5. $(xy - c)(x^2 - y^2 - c) = 0$ 6. $(y - e^x - c)(y - e^{-x} - c) = 0$
 7. $(y - \frac{1}{2}x^2 + c)(y + x + ce^{-x} - 1) = 0$ 8. $(y^2 + x^3 - c)(y^2 + \frac{1}{2}x^2 - c) = 0$.

11.15. EQUATIONS SOLVABLE FOR y

If the equation is solvable for y , we can express y explicitly in terms of x and p . Thus, the equations of this type can be put as $y = f(x, p)$... (1)

Differentiating (1) w.r.t. x , we get $\frac{dy}{dx} = p = F\left(x, p, \frac{dp}{dx}\right)$... (2)

Equation (2) is a differential equation of first order in p and x .

Suppose the solution of (2) is $\phi(x, p, c) = 0$... (3)

Now elimination of p from (1) and (3) gives the required solution.

If p cannot be easily eliminated, then we solve equations (1) and (3) for x and y to get

$$x = \phi_1(p, c), y = \phi_2(p, c)$$

These two relations together constitute the solution of the given equation with p as parameter.

ILLUSTRATIVE EXAMPLES

Example 1. Solve $y + px = x^4p^2$.

Sol. Given equation is $y = -px + x^4p^2$... (1)

Differentiating both sides w.r.t. x ,

$$\frac{dy}{dx} = p = -p - x \frac{dp}{dx} + 4x^3p^2 + 2x^4p \frac{dp}{dx}$$

or $2p + x \frac{dp}{dx} - 2px^3 \left(2p + x \frac{dp}{dx}\right) = 0$

or $\left(2p + x \frac{dp}{dx}\right)(1 - 2px^3) = 0$

Discarding the factor $(1 - 2px^3)$, we have $2p + x \frac{dp}{dx} = 0$ or $\frac{dp}{p} + 2 \frac{dx}{x} = 0$

Integrating $\log p + 2 \log x = \log c$

or $px^2 = c \Rightarrow p = \frac{c}{x^2}$.

Putting this value of p in (1), we have $y = -\frac{c}{x} + c^2$.

Example 2. Solve $y = 2px - p^2$.

Sol. The given equation is $y = 2px - p^2$... (1)

Differentiating both sides w.r.t. x , $\frac{dy}{dx} = p = 2p + 2x \frac{dp}{dx} - 2p \frac{dp}{dx}$

or $p + (2x - 2p) \frac{dp}{dx} = 0$

or
$$p \frac{dx}{dp} + 2x - 2p = 0$$

or
$$\frac{dx}{dp} + \frac{2}{p} x = 2 \quad \dots(2)$$

which is a linear equation.

$$\text{I.F.} = e^{\int \frac{2}{p} dp} = e^{2 \log p} = p^2$$

$\therefore$ The solution of (2) is
$$xp^2 = \int 2p^2 dp + c$$

or
$$xp^2 = \frac{2}{3} p^3 + c \quad \text{or} \quad x = \frac{2}{3} p + cp^{-2} \quad \dots(3)$$

Putting this value of x in (1), we have
$$y = 2p \left(\frac{2}{3} p + cp^{-2} \right) - p^2$$

or
$$y = \frac{1}{3} p^2 + 2cp^{-1}. \quad \dots(4)$$

Equations (3) and (4) together constitute the general solution of (1).

TEST YOUR KNOWLEDGE

Solve the following equations :

1. $xp^2 - 2yp + ax = 0$

2. $y - 2px = \tan^{-1}(xp^2)$

3. $16x^2 + 2p^2y - p^3x = 0$

4. $y = x + 2 \tan^{-1} p$

5. $y = 3x + \log p$

6. $x - yp = ap^2$

7. $x^2 \left(\frac{dy}{dx} \right)^4 + 2x \frac{dy}{dx} - y = 0.$

Answers

1. $2y = cx^2 + \frac{a}{c}$

2. $y = 2\sqrt{cx} + \tan^{-1} c$

3. $16 + 2c^2y - c^3x^2 = 0$

4. $x + c = \log \frac{p-1}{\sqrt{p^2+1}} - \tan^{-1} p$, with the given relation

5. $y = 3x + \log \frac{3}{1 - ce^{3x}}$

6. $x = \frac{p}{\sqrt{1-p^2}} (c + a \sin^{-1} p), y = \frac{1}{\sqrt{1-p^2}} (c + a \sin^{-1} p) - ap.$

7. $y = c^2 + 2\sqrt{cx}.$

11.16. EQUATIONS SOLVABLE FOR x

If the equation is solvable for x , we can express x explicitly in terms of y and p . Thus, the equations of this type can be put as $x = f(y, p)$...(1)

Differentiating (1) w.r.t. y , we get
$$\frac{dx}{dy} = \frac{1}{p} = F \left(y, p, \frac{dp}{dy} \right) \quad \dots(2)$$

Equation (2) is a differential equation of first order in p and y .

Suppose the solution of (2) is $\phi(y, p, c) = 0$... (3)

Now elimination of p from (1) and (3) gives the required solution.

If p cannot be easily eliminated, then we solve equations (1) and (3) for x and y to get

$$x = \phi_1(p, c), y = \phi_2(p, c)$$

These two relations together constitute the solution of the given equation with p as parameter.

ILLUSTRATIVE EXAMPLES

Example 1. Solve $y = 2px + y^2p^3$.

Sol. Solving for x , we have $x = \frac{1}{2} \left(\frac{y}{p} - y^2p^2 \right)$

Differentiating both sides w.r.t. y

$$\frac{dx}{dy} = \frac{1}{p} = \frac{1}{2} \left(\frac{1}{p} - \frac{y}{p^2} \cdot \frac{dp}{dy} - 2yp^2 - 2y^2p \frac{dp}{dy} \right)$$

or
$$2p = p - y \frac{dp}{dy} - 2yp^4 - 2y^2p^3 \frac{dp}{dy}$$

or
$$p + 2yp^4 + y \frac{dp}{dy} + 2y^2p^3 \frac{dp}{dy} = 0$$

or
$$\left(p + y \frac{dp}{dy} \right) (1 + 2yp^3) = 0$$

Discarding the factor $(1 + 2yp^3)$, we have

$$p + y \frac{dp}{dy} = 0 \quad \text{or} \quad \frac{dy}{y} + \frac{dp}{p} = 0$$

Integrating $\log y + \log p = \log c$ or $py = c$ or $p = \frac{c}{y}$

Putting this value of p in the given equation, we have

$$y = \frac{2cx}{y} + \frac{c^3}{y} \quad \text{or} \quad y^2 = 2cx + c^3$$

Example 2. Solve $p = \tan \left(x - \frac{p}{1+p^2} \right)$.

Sol. Solving for x , we have $x = \tan^{-1} p + \frac{p}{1+p^2}$... (1)

Differentiating both sides w.r.t. y ,

$$\frac{dx}{dy} = \frac{1}{p} = \frac{1}{1+p^2} \cdot \frac{dp}{dy} + \frac{(1+p^2) - 2p^2}{(1+p^2)^2} \cdot \frac{dp}{dy}$$

or
$$\frac{1}{p} = \frac{2(1+p^2) - 2p^2}{(1+p^2)^2} \frac{dp}{dy} \quad \text{or} \quad dy = \frac{2p}{(1+p^2)^2} dp$$

Integrating
$$y = c - \frac{1}{1+p^2} \quad \dots(2)$$

Equations (1) and (2) together constitute the general solution.

TEST YOUR KNOWLEDGE

Solve the following equations :

1. $y = 3px + 6p^2y^2$
2. $y = 2px + p^2y$
3. $p^3 - 4xyp + 8y^2 = 0$
4. $y^2 \log y = xyp + p^2$.

Answers

1. $y^3 = 3cx + 6c^2$
2. $y^2 = 2cy + c^2$
3. $64y = c(c - 4x)^2$
4. $\log y = cx + c^2$.

11.17. CLAIRAUT'S EQUATION

An equation of the form $y = px + f(p)$...(1)

is known as Clairaut's equation.

Differentiating (1) w.r.t. x , we get

$$p = p + x \frac{dp}{dx} + f'(p) \frac{dp}{dx} \quad \text{or} \quad [x + f'(p)] \frac{dp}{dx} = 0$$

Discarding the factor $[x + f'(p)]$, we have $\frac{dp}{dx} = 0$

Integrating $p = c$

Putting $p = c$ in (1), the required solution is $p = cx + f(c)$

Thus, the solution of Clairaut's equation is obtained by writing c for p .

ILLUSTRATIVE EXAMPLES

Example 1. Solve $(y - px)(p - 1) = p$.

Sol. The given equation can be written as

$$y - px = \frac{p}{p-1} \quad \text{or} \quad y = px + \frac{p}{p-1}$$

This is of Clairaut's form. Hence putting c for p , the solution is $y = cx + \frac{c}{c-1}$.

Note. Many differential equations can be reduced to Clairaut's form by suitably changing the variables.

Example 2. Solve $e^{4x}(p-1) + e^{2y}p^2 = 0$.

Sol. [In problems involving e^{lx} and e^{my} , put $X = e^{kx}$ and $Y = e^{ky}$, where k is the H.C.F. of l and m].

Put $X = e^{2x}$ and $Y = e^{2y}$
 so that $dX = 2e^{2x} dx$ and $dY = 2e^{2y} dy$

$$\therefore p = \frac{dy}{dx} = \frac{e^{2x}}{e^{2y}} \frac{dY}{dX} = \frac{X}{Y} P, \text{ where } P = \frac{dY}{dX}$$

The given equation becomes $X^2 \left(\frac{X}{Y} P - 1 \right) + Y \cdot \frac{X^2}{Y^2} P^2 = 0$

or $XP - Y + P^2 = 0$ or $Y = PX + P^2$ which is of Clairaut's form

$\therefore$ Its solution is $Y = cX + c^2$ and hence $e^{2y} = ce^{2x} + c^2$.

Example 3. Solve $(px - y)(py + x) = 2p$.

Sol. Put $X = x^2$ and $Y = y^2$ so that $dX = 2x dx$ and $dY = 2y dy$

$$\therefore p = \frac{dy}{dx} = \frac{x}{y} \frac{dY}{dX} = \frac{\sqrt{X}}{\sqrt{Y}} P, \text{ where } P = \frac{dY}{dX}$$

The given equation becomes

$$\left(\frac{\sqrt{X}}{\sqrt{Y}} P \cdot \sqrt{X} - \sqrt{Y} \right) \left(\frac{\sqrt{X}}{\sqrt{Y}} P \cdot \sqrt{Y} + \sqrt{X} \right) = 2 \frac{\sqrt{X}}{\sqrt{Y}} P$$

or $(PX - Y)(P + 1) = 2P$

or $Y = PX - \frac{2P}{P+1}$ which is of Clairaut's form.

$\therefore$ Its solution is $Y = cX - \frac{2c}{c+1}$ and hence $y^2 = cx^2 - \frac{2c}{c+1}$.

TEST YOUR KNOWLEDGE

Solve the following equations :

1. $y = xp + \frac{a}{p}$

2. $y = px + \sqrt{a^2 p^2 + b^2}$

3. $p = \log (px - y)$

4. $p = \sin (y - px)$

5. $p^2(x^2 - 1) - 2pxy + y^2 - 1 = 0$

6. $e^{3x}(p - 1) + p^3 e^{2y} = 0$

7. $x^2(y - px) = yp^2$

8. $(y + px)^2 = x^2 p$.

Answers

1. $y = cx + \frac{a}{c}$

2. $y = cx + \sqrt{a^2 c^2 + b^2}$

3. $y = cx - e^c$

4. $y = cx + \sin^{-1} c$

5. $(y - cx)^2 = 1 + c^2$

6. $e^y = ce^x + c^2$

7. $y^2 = cx^2 + c^2$

8. $xy = cx - c^2$.